

Bryan Loh – Security Engineer

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PROFILE

Software engineer with 2 years of full-stack experience at DBS Bank and Tata Consultancy Services, currently completing a Master of IT (Cyber Security Engineering) at UNSW on a Scientia Scholarship. Interested in work that spans software, security, and infrastructure with hands-on exposure across all three through professional projects and self-driven learning.

SKILLS

Languages: Java · Python · JavaScript · SQL · Bash · HTML/CSS

Backend & Frameworks: Spring Boot · Flask · Express · Node.js · REST API Design · PostgreSQL · MongoDB · JUnit/Jest

Frontend: React · Next.js · Astro · StencilJS

DevOps & Cloud: Docker · AWS EC2 · AWS S3 · CloudFront · Route53 · ACM · Terraform · Digital Ocean · Caddy · Nginx · GitHub Actions · Git · Maven/Gradle

Security: Burp Suite · Snort · ELK Stack · nmap · Wireshark · ffuf · sqlmap · Metasploit · OWASP Top 10

Concepts: Authentication & Authorisation · Web Application Vulnerabilities · TCP/IP Networking · Linux Fundamentals · Agile/Scrum

WORK EXPERIENCE

Software Engineer — DBS Bank | Singapore

Jul 2023 – Jul 2024

- Architected and implemented backend services in Java and Spring Boot for a bank-wide change management workflow system, designing REST APIs and data models to support auditability and regulatory compliance
- Enhanced existing REST APIs on a high-traffic banking platform, optimising backend processing and database interactions to improve customer information update efficiency by ~50%
- Managed project timelines and proactively communicated requirements across cross-functional stakeholders in an agile environment

Software Engineer — DBS Bank & Tata Consultancy Services | Singapore

Sep 2022 – Jun 2023

- Developed and maintained Spring Boot backend services for an enterprise customer loyalty application, translating business logic into production-grade services
- Led the migration of React components to web components using StencilJS, enabling cross-framework reuse and tripling maintainability
- Mentored colleagues transitioning from non-technical backgrounds, providing structured guidance to accelerate their ramp-up

Senior Resolution Generalist — IKEA Australia | Sydney

Sep 2024 – Present

- Discovered and responsibly disclosed an IDOR vulnerability in a third-party platform that exposed PII (name, email, phone, address) of 13,000+ customers to any authenticated user

PROJECTS

AWS Static Site Hosting & IaC (Terraform) | Personal – 2026

AWS S3 · CloudFront · Route53 · ACM · Terraform

- Hosted a static site on Amazon S3 served globally via a CloudFront CDN, with a TLS certificate from AWS Certificate Manager (ACM) and DNS managed in Route53 — served over HTTPS on a bryanloh.dev subdomain
- Automated the entire hosting stack (S3, CloudFront, Route53, ACM) as code with Terraform for repeatable, version-controlled provisioning

bryanloh.dev — Personal Site with Full CI/CD Pipeline | Personal – 2026

Astro · Vite · GitHub Actions · Gitleaks · Digital Ocean · Caddy

- Built and deployed a static site on a Digital Ocean droplet with a complete GitHub Actions CI/CD pipeline — automated linting, Gitleaks secret scanning, and security checks on every PR, with automatic deployment on merge to main
- Configured pre-commit hooks, DNS, and Caddy as a public-facing reverse proxy — full ownership from domain registration through to production

Home Network Intrusion Detection System | Personal – 2025

- Engineered a real-time IDS on a Raspberry Pi using Snort, integrated with the ELK Stack for centralised log ingestion, threat analysis, and custom Kibana dashboards
- 1st Place** — SECedu Conference Poster Competition (Jun 2025)

Spotify Remote Queue — Full-Stack Web App | Personal – 2023

React · Spring Boot · PostgreSQL · AWS EC2 · Nginx · OAuth 2.0

- Designed, built, and shipped a publicly accessible full-stack app end-to-end, letting Spotify users generate shareable links so others can add to their live queue in real time without a Spotify account
- Deployed and operated on AWS EC2 with Nginx reverse proxy, managed PostgreSQL, and OAuth 2.0 integration with the Spotify Web API

Multi-VLAN Home Network | Personal – Dec 2025
OpenWRT · Raspberry Pi · VLANs · Firewall Policy

- Designed and deployed a segmented home network (Home / Guest / IoT VLANs) on a Raspberry Pi running OpenWRT, enforcing inter-VLAN firewall rules to block IoT lateral movement and restrict Guest traffic to internet-only — applying enterprise-grade segmentation patterns to a home environment

Web Application Security Assessment | Academic – UNSW – 2025
Burp Suite · Python · Bash · OWASP Top 10

- Conducted structured penetration testing across 8 web applications, identifying 20+ vulnerabilities (XSS, SQL injection, authentication bypass) and producing technical and executive vulnerability reports with risk ratings and remediation guidance
- Built reusable reconnaissance and enumeration scripts in Python and Bash, reducing manual testing effort by 50%

AI Hackathons — UNIHACK & Mistral AI Worldwide | Mar 2026
Next.js · React · Flask · MongoDB · Google Gemini API · Mistral API

- Shipped two end-to-end LLM-integrated products within hackathon timeframes: a fridge digital twin with AI-powered receipt scanning and recipe generation (Google Gemini), and a hiring platform with dynamic AI-driven interviews (Mistral)

EDUCATION

Master of Information Technology (Cyber Security Engineering) – UNSW Sydney Feb 2025 – Dec 2026
WAM: 87/100 (High Distinction) · Scientia Scholarship
Key courses: Securing Fixed & Wireless Networks · Computer Networks · Database Systems · Digital Forensics · Extended Web Security · Security Assessment · Data Structures & Algorithms

Bachelor of Engineering (Mechanical) — NUS 2019 – 2022
GPA: 4.90 / 5.00 - First Class Honours
Developed structured problem-solving and systems thinking through rigorous engineering training — directly applicable to software architecture and root cause analysis

CERTIFICATIONS & AWARDS

Certified in Cybersecurity (CC) – ISC2	Jul 2024
Practical Ethical Hacking – Complete Course – TCM Security	Oct 2024
AWS Certified Cloud Practitioner – AWS	Jun 2026
1st Place – Poster Competition – SECedu Conference	Jun 2025
Top 4 – Ninja Night CTF (web exploitation) – SecTalks Sydney	May 2026
Top 10 / 60 Teams – Jingle CTF – ACUCyS	Dec 2025